

Investigation into the parameters that manipulate the simulation of the advection of 3D fluids

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Abstract: the purpose of the project was to identify the properties of the fluid field and simulate the advection using a methodology that demonstrated the power of customised parameters

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1. Introduction

1.1 Background

This project investigated and analysed the techniques and methodologies available for the simulation of fluids in 3D animation. The research has been implemented into the project with the creation of a system that allowed to control the simulation of a different range of fluids, focused on one of the properties of fluids. The results of the simulations have been evaluated and used to create a guideline that shows the different criteria and requirements, necessary to control the fluid behaviour in a specific scenario.

The literature review covered topics such as the description of the general properties of fluids, the governing laws that control their behaviours, how fluids are generated in computer graphics and the variety of techniques and solvers that are available to simulate fluids. The simulation of fluids is a process that requires to use methods that need to be solved by a computer program utilising tools termed solver. The project has been created using a node-based software, that allowed to develop a fluid solver with customised parameters.

1.2 Aims and objectives

The aim of this project is to study how fluids are simulated in 3D animation to identify which are the parameters and tools that control their behaviour and allow their manipulation.

To achieve the purpose of the project the following objectives have been pursued:

- Analysis of the governing laws that influence the fluid dynamics in the physical world;
- Research of the methods that allow to study the fluid field;
- Investigate the solvers available to simulate fluids in virtual environments;
- Inspect the structure of volumetric simulations and the anatomy of fluid solvers;
- State the properties that characterise fluids;
- Simulate one specific property of fluids through a range of simulations;
- Compare and examine the results of the simulations;
- Create a guideline that allows to evaluate the results.

1.3 Overview of the chapters

The following report is divided in four main sections. The first section discusses the literature review which includes the understanding of the principles of fluids, past work on fluid simulations and how they are used in virtual environments. This section concludes, stating the properties of the fluid field and which of the key factors are useful for the simulations of fluids in animation.

In the second section, experimentations have been made to understand which technologies are available to simulate fluids and how theories are applied in the current software packages used in animation. After the experiments, one tool and one methodology have been chosen to generate one specific property of fluids through a range of simulations.

Therefore, in the third section, the workflow to create the simulations has been explained. The decisions made have been justified according to the research and the experimentations.

The last section gathered the results of the simulations and it also compared the data in order to create a guideline that allowed to identify which are the criteria used to manipulate one specific property of fluids.

2. Literature Review

2.1 Approach to literature searching

Research has been conducted using online databases such as Science Direct and ACM Digital Library (Association of Computing Machinery). The selected papers and academic journals that helped with the argumentation of the report are based on proceedings of the annual conference SIGGRAPH (Special Interest Group on GRAPHics and Interactive Techniques) annual conference. Other fundamental resources for the achievement of the project were the online tutorials provided by the Side Effects Software Inc. and books about fluid dynamics such as Fluid Simulation for Computer Graphics by Robert Bridson (Bridson, 2008).

2.2 Principles of fluid dynamics

A fluid is a substance which cannot resist shear forces without moving. Fluids are usually classified as liquid or gases (Hughes & Brighton, 1999). They behave similarly under action of forces, but their properties differ. The most valuable parameters used for describing a fluid in the real world are density, pressure, flow and viscosity (Ferziger & Peric', 2002):

- The density (ρ) is the mass per unit volume of a fluid. In general, the density of a fluid decreases with the increase of the temperature (T) and it increases with the increment of the pressure (P). The density varies in every fluid and the ideal gas equation is used to find density of any fluid if pressure and temperature are known:

$$P = \rho RT$$
 Equation of state of gas where R is the gas constant value ($8.3144598(48) \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$)
- The pressure is defined as the amount of force applied per area $P: \frac{F}{A}$. To study the internal pressure of a fluid the velocity and acceleration need to be null.
- A fluid flow can be laminar or turbulent. The flow is laminar if each particle of the fluid follows the path of its preceding particle and the particles move smoothly in layers one over the other. On the contrary, the flow is turbulent if the particles do not follow any regular pattern while flowing, and individual particles cross one another. The Reynolds number (Re) is used to determine the flow of a fluid. If Re is under 2300, the fluid flow is laminar, if Re is more than 4000 it is turbulent. If it is between 2300 and 4000, the fluid flow is unstable.
- The viscosity determines the amount of resistance in a fluid to shear stress, called internal friction. In a liquid, viscosity decreases with the increasing temperature while in a gas it increases.

Mathematically, the state of a fluid with constant temperature and constant density is described by a vector field that assigns a velocity vector to each point in the space. The model that describes the behaviour of the field in time is given by Claude- Louis Navier in 1821 and George Gabriel Stokes in 1845 where they elaborate the same conclusions. Their general analytical resolution is still one of the unresolved problems of modern mathematics, but they are fundamental to describe the motion of fluids and to determinate the properties of the fluid field (Quarteroni, 2008).

The Navier- Stokes equations are formulated based on the principles of conservation of mass and energy, these will be examined, referring to a generic incompressible fluid. A fluid is considered incompressible when the density is constant. To better understand the equations some elements, need to be identified (see Table 1.).

p	pressure
ν	viscosity
ρ	density
f	external forces
u	Velocity vectors in the three coordinates x, y, z (u,v,w)
∇	vector differential operator that denote the gradient of the scalar field
Δ	Laplace operator

Table 1. Equations Nomenclature

- First equation: based on the law conservation of the mass it explains that inside a defined amount of space the volume is always constant.

$$\frac{\partial \bar{u}}{\partial t} + (\bar{u} \cdot \nabla) \bar{u} = \nu \Delta \bar{u} - \frac{1}{\rho} \nabla p + \frac{1}{\rho} \bar{f}$$

- Second equation: based on the law of the conservation of energy it describes the velocity profile of the flow.

$$\text{div}(\rho \bar{u}) = 0$$

2.3 Computational fluid dynamics

The Navier- Stokes equations can be solved by computers, and used in Computational Fluid Dynamics (CFD) which is a discipline that has the aim to simulate the fluid flow to obtain physical insights and quantitative predictions (Hall, 2015).

The first CG fluid experimentations can be associated at the Los Alamos National Laboratory where Francis Harlow and the T3 Group researched in the field since 1953 and formed the core of the CFD.

The aim of the group was to develop numerical techniques that would aid the USA's national defence system. Their techniques improved with the evolution of high speed computers (Harlow, 2004).

The studies made in CFD field were important, but the principles were not used for large scale simulations and visual effects. This has been a challenge before the thesis of Foster and Metaxas (Foster & Metaxas, 1996) and later of Stam (Stam, 1999) which with their studies they allowed the future generations to create applications able to simulate and control fluids:

- Foster and Metaxas presented a method to animate liquids solving the Navier-Stokes equations, providing a range of behaviour and effects not possible with previous computer graphics fluid models. Their fluid model demonstrated effects such as wave effects as reflection, refraction and diffraction and rotational effects as vorticity and splashing. Their model was created whilst thinking of a fluid simulator for graphics applications that would apply the correct conditions automatically and satisfy the needs of the animator. They concluded that to minimize computation costs, boundaries between the liquids and the surroundings needed to be incorporated in the model. Also, they demonstrate that the user of the system needs to be able to specify the geometry and apply external control to the system (Foster & Metaxas, 1996).
- Stam introduced an innovative point defining a method to stabilize unstable simulations. This study has enabled the creation of real-time simulations in which the user could interact during the animation, applying external forces. Before Stam, it was believed that physical fluid models were too expensive to allow real-time interaction because previous models used unstable schemes to solve the physical equations of the fluid. This changed after the formulation of an algorithm different from the one created by Foster and Metaxas (Stam, 1999).

2.4 Fluid solvers and techniques

The properties of a fluid field are studied following two methods of investigation, Eulerian and Lagrangian (Hughes & Brighton, 1999):

- The Eulerian method is focused on analysing a fixed volume in the space into a three-dimensional grid where only the volume inside the cell is being observed when the fluid moves inside. The velocities are calculated at each point on that grid, while the grid is not moving with the fluid (see Fig.1) (Thacker, 2010).

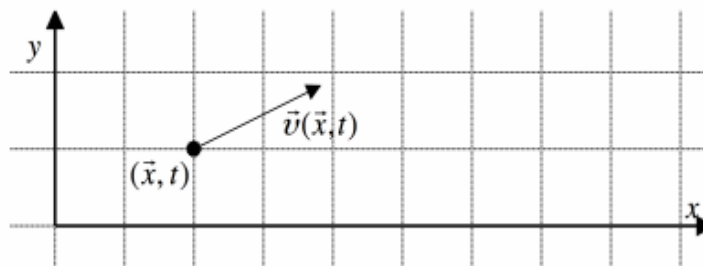


Fig.1 Eulerian system where only the cells in one fixed grid determines the velocity of the field

- The Lagrangian system is based on the study of the movement of every single particle. The motion of a particle is being followed at each point of the flow, which determinates the properties of the field (see. Fig 2) (Thacker, 2010).

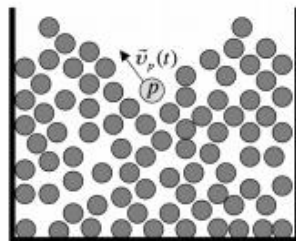


Fig.2 Lagrangian system where the particle determines the velocity of the field (Ertekin, 2015)

Based on these two methods fluid solvers are created and adapted to simulate fluids. The most common techniques are the following:

- The Marker and Cell (MAC Grid): the MAC technique has a Eulerian approach where there is a field inside a grid. A volume of fluid is divided in cells, every cell has its own finite volume, the pressure is defined at the centre of every cell and the velocity is defined at the centre of every face (see Fig.3) (Harlow, 2004).

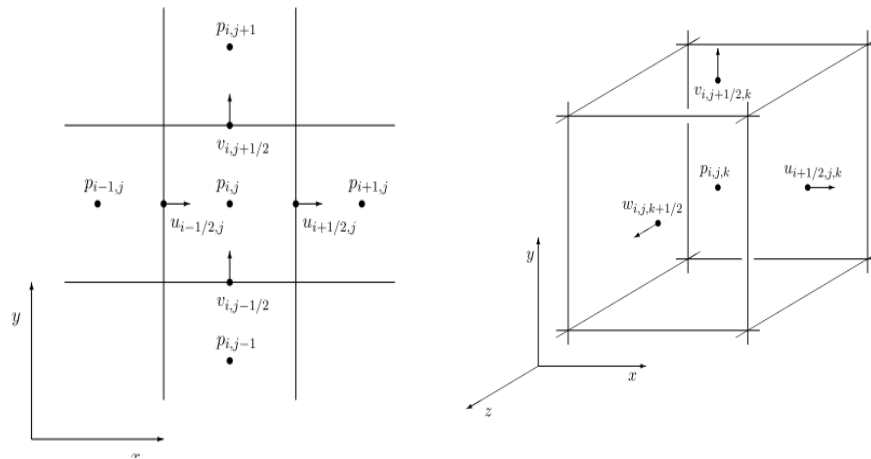


Fig.3 Two Dimensional MAC Grid and one cell from the three-dimensional MAC grid (Harlow, 2004)

- Smoothed Particle Hydrodynamics (SPH): The smoothed particle hydrodynamics is a Lagrangian mesh-free method initially used in astronomy by Gingold and J.J. Monaghan (Monaghan, 1992). Desbrun and Gascuel (Desbrun and Gascuel, 1996) were the first to apply SPH to simulate surfaces. After them, Muller (Müller, Charypar and Gross, 2003) implemented the SPH technique with the Navier-Stokes equations. An evolution of the SPH method is the Position Based Fluid (PBD) that shows more stability for real time applications (see Fig.4).

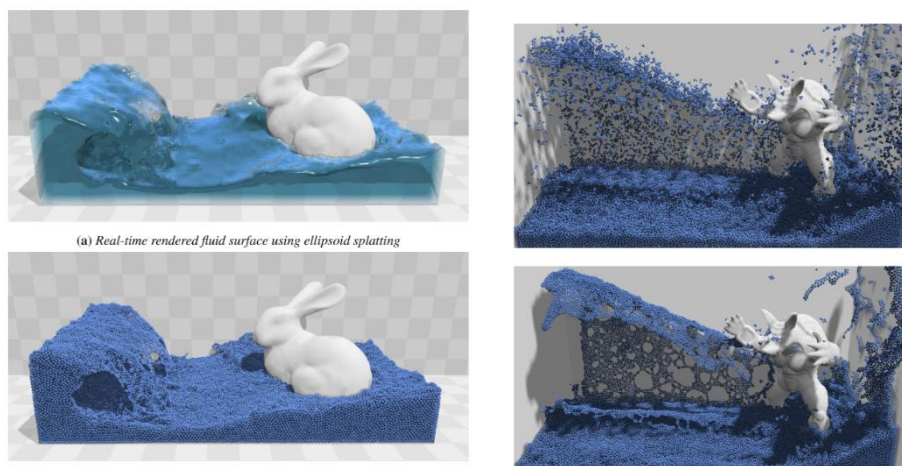


Fig.4 Application of SPH technique by Muller (Bridson and Müller-Fischer, 2007)

- Implicit Particles (PIC/FLIP and APIC): those techniques have a hybrid approach, that mixes the characteristics of the techniques analysed before. The Particle- In- Cell (PIC) method was introduced by Harlow in 1963, which presents a Eulerian set of cells along with a Lagrangian particle marker that moves through the cells (Harlow, 2004). The Fluid Implicit Particles (FLIP) technique was showed by Zhu and Bridson (Bridson & Zhu, 2005) where they take the strengths of particles and grids, offering improved flexibility and efficiency. The Affine Particle In Cell (APIC) technique is designed to retain the stability of the characteristics of PIC, without suffering from the noise of FLIP (Jiang et al., 2015). PIC and FLIP techniques are powerful but present some limitations, PIC is stable, but highly dissipative, while FLIP is designed to remove this dissipation, but it is unstable (see Fig.5) (Jiang et al., 2015).

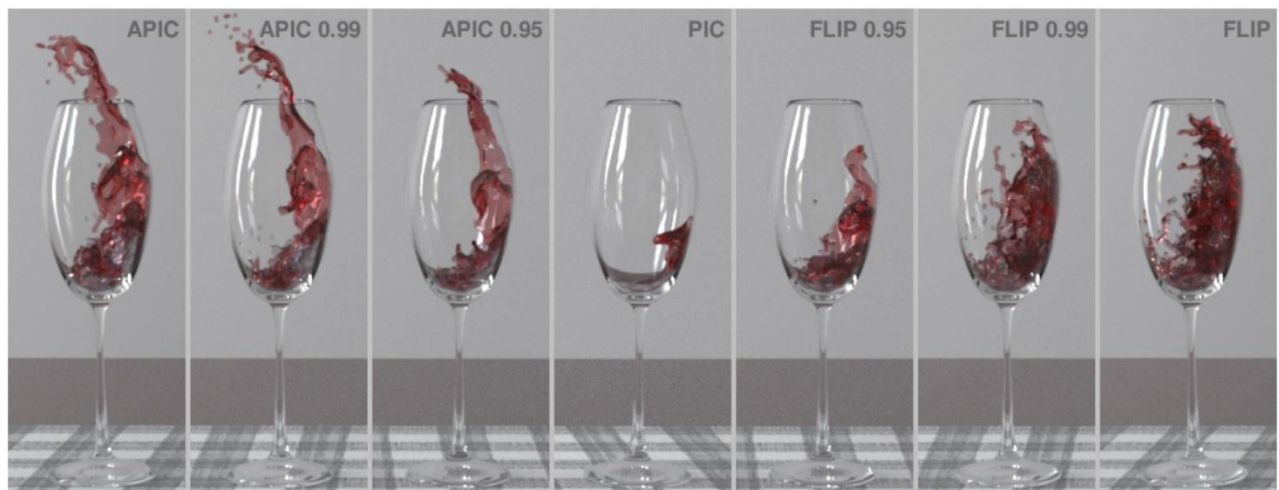


Fig. 5 Testing of the Affine Particle in Cell technique (Jiang et al., 2015)

2.5 Properties of the fluid field

The Navier-Stokes equations are the main reference in understanding the properties of the fluid field, if the first equation is divided into parts it affirms that a change in the velocity depends on four main factors: advection, diffusion, pressure and body forces (see fig.6):

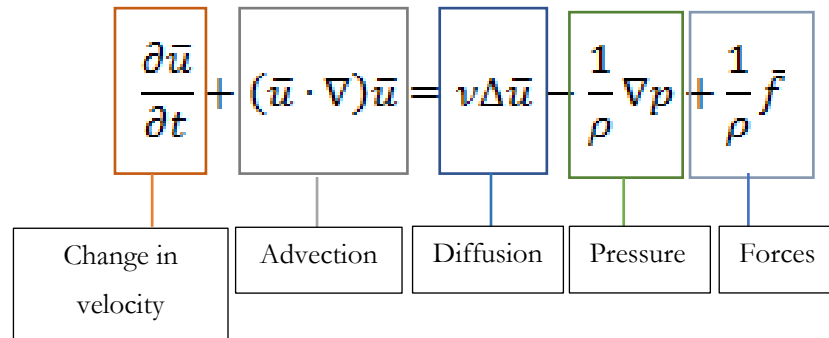


Fig.6: Breaking down of the fluid field properties from the Navier-Stokes equations

- The advection defines how molecules and particles are transported and how they move in a given velocity field (Bridson & Muller-Fischer, 2007). It is common to notice advection in phenomena such as the water cycle and meteorology events where heat, humidity and salinity are transported by the movement of the surrounding fluid.
- The diffusion defines how a quantity in a cell, exchanges molecules with the nearest cells. Viscosity is a consequence of diffusion which defines the amount of friction between particles. Based on the viscosity, fluids are classified in Newtonian and non-Newtonian. Newtonian fluids are the ones that have constant or predictable viscosity. For example, water has predictable viscosity regardless of the forces acting on it. Also, the honey is considered Newtonian because even if the viscosity changes in response to temperature and pressure the results are predictable. Non-Newtonian fluids instead have unpredictable responses to applied forces that create viscosity changes, example of this fluid are paint and blood.
- The pressure is responsible in the creation of a gradient, forces are created from the imbalance of the gradient when the particles move in direction of largest change in pressure (Bridson & Zhu, 2005). Related to the pressure there is the density which is the mass divided for the volume.
- External forces such as wind and gravity create changes in the behaviour of the fluids that generates phenomena such as vorticity and acceleration.

2.6 Application of fluid simulations

Computational Fluid Dynamics is a discipline applied in many fields, the main are the entertainment industry and the scientific disciplines.

The animation of fluids like water, smoke, and fire by physics-based simulation is increasingly important in the movie industry and makes an impact in real-time games (Bridson & Muller-Fischer, 2007).

Recently, water came to life for the production of the animated movie Moana (Moana, 2016) where water was presented as a character. This was achieved using a proprietary fluid APIC solver (Jiang et al, 2015) named Splash (Frost, et al., 2017).

Before the existence of computer software packages which were able to control the fluids, water effects were simulated using traditional frame by frame animation and 2D paintings. The first 3D digital water effects were created for the Pseudopod creature in The Abyss (The Abyss, 1989). The effect was created investigating and developing digital technology previously unheard accelerating the evolution of 3D computer graphics (Lucasfilm, 2017). Simulations are becoming more challenging because the expectations are higher due to the evolution of the technologies. For example, the latest Pirates of the Caribbean (Pirates of the Caribbean: Dead Men Tell No Tales, 2017) created highly believable ocean and water effects that interact with computer generated ships and characters (Hopper & Wolter, 2017).

The aim of fluid simulation in the visual effects industry, as opposed to the physics community, is not to make reliable predictions, instead, the main purpose is to capture the look of the fluid motion and make it believable and interesting for the audience (Lagergren, 2010). Existing 3D software packages often limited the 3D artist control, that is why most of the VFX companies use customised tools, in-house software or customised versions of commercial software to quickly achieve their requirements (Bodenstein, 2012). Example of independent tools that VFX companies use are PhysBAM, Plume, FSI from Digital Domain, Matter Matterhorn Engine Disney and Naiad's solver custom fluid simulation tools used at Double Negative.

Other fields where CFD principles are applied, are for example the medical for blood simulations, engineering investigations such as aerospace, hydraulics, chemical processing, meteorological phenomena. However, because of their different purpose these simulations follow a different approach by not using the traditional fluid simulator software packages however using toolbox for the development of numerical solvers like OpenFoam (CFD Direct, 2015).

2.7 Summary of literature review

The literature review covered different aspects related to the dynamics of fluids and their simulations. In fluid dynamics simulation, one relevant principle is the Navier-Stokes Equations that are used to give values to the fluids in the real world and used by machinery to create fluid solver in the virtual world. Analysing the different methodologies and property of the fluids, it can be summarized that there is a variety of techniques that can be applied to simulate fluids and that the approach is different depending on the results desired.

The research made in this field are multitudinous and it is an active topic that draw the attention of many physicians and artists, and it is still in continuous evolution.

Most of the researchers have mathematical insights, always trying to improve the current techniques and find new schemes to propose alternative algorithms that could be applied in future developments. Therefore, progresses in computational fluid dynamics often cannot be directly applied to computer graphics because they have different goals.

In the animation field, to produce certain special fluid effects animators have to manipulate numerous parameters repeatedly, which can result laborious and time consuming. Furthermore, the existing methods can be expensive and slow to use, and long simulation time and high memory are required to simulate fluids. In the current animation industry, it is not only important to be in control of the fluid but also on the motion and look that the visual artist want to create.

There is no standard approach that allows to manipulate fluids because they can be unpredictable, although it is possible to establish some parameters there are commonly manipulated. This has been explored in the following sections.

3. Experimentations

3.1 Background of the technologies

After the research, some technologies have been reviewed to see how theories are applied in the software packages that allow to simulate fluids. The purpose of the experimentations was to compare fluid solvers and fluid properties. The simulation packages used were RealFlow and the Autodesk 3Ds Max and Maya. RealFlow is a simulation software with a focus on liquids, it has a variety of solvers available, the main engine in the software are Dyverso and Hybrido. Dyverso applies two Lagrangian methods SPH and PBD (see section 2.4). Even though SPH is an older technology it provides accurate information, PBD is fast and stable. Dyverso is mainly used for small and mid-scaled projects as well as to create rich fluids with fine structures in order for it to be used when high level of detail is required. Hybrido instead uses, like the name suggests, hybrid technologies between grids and particles which is the FLIP method. In this FLIP the particles are used to carry the fluid's velocity information and during the simulation, the velocity is transferred to the grid. Hybrido is mainly used for large simulations that require considerable amount of water volumes (Next Limit Technologies, 2017).

Autodesk 3Ds Max recently updated the fluid feature which is an implementation of the Bifröst simulation engine. Bifröst is a framework that uses the FLIP solver. It is possible to modify the setting of the solver by changing the voxel of the simulation, the same Bifrost technology is used in Maya (Autodesk, 2017). Those technologies have been used to do some basic simulations and become familiar with the volumetric environments.

3.2 Experimental simulations

The Realflow software has been used to experiment with the Lagrangian techniques using Dyverso PBD and Dyverso SPH, analysing the same simulations and changing only the type of solver (see Fig9).

From the experiments using those techniques, it is resulted that the manipulation of the sub steps is an important factor that needs to be considered during the simulations of fluids. The sub-steps are the unit that calculates how the fluid behave in time. Increase the time steps is a good solution to have accurate results but at the cost of longer simulations (Next Limit Technologies, 2017). Dyverso SPH requires a higher number of sub steps while the Dyverso PBD is stable even with less number of sub steps, faster and more accurate.

The 3Ds Max's fluid solver has been used to experiment with the Eulerian solver (see Fig.10). The fluid solver works by simulating particles through an adaptive grid of voxels. From this simulation it resulted to have highly detailed fluid, smaller voxels are better to capture details, the simulation run faster if the voxel size is bigger although this would make the results more inaccurate. The default voxel size is set to 0.5 units which produces a high-resolution fluid, the smaller the voxel the more detailed the fluid results.

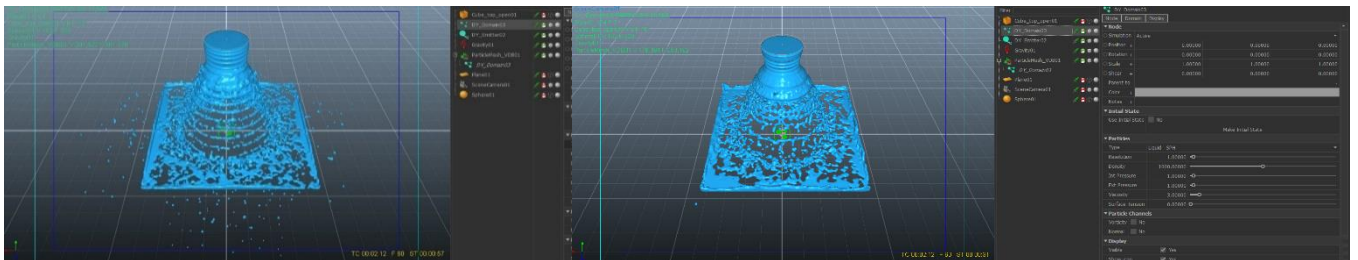


Fig. 9. Solver Particle based, Frame 60 SPH vs PBD

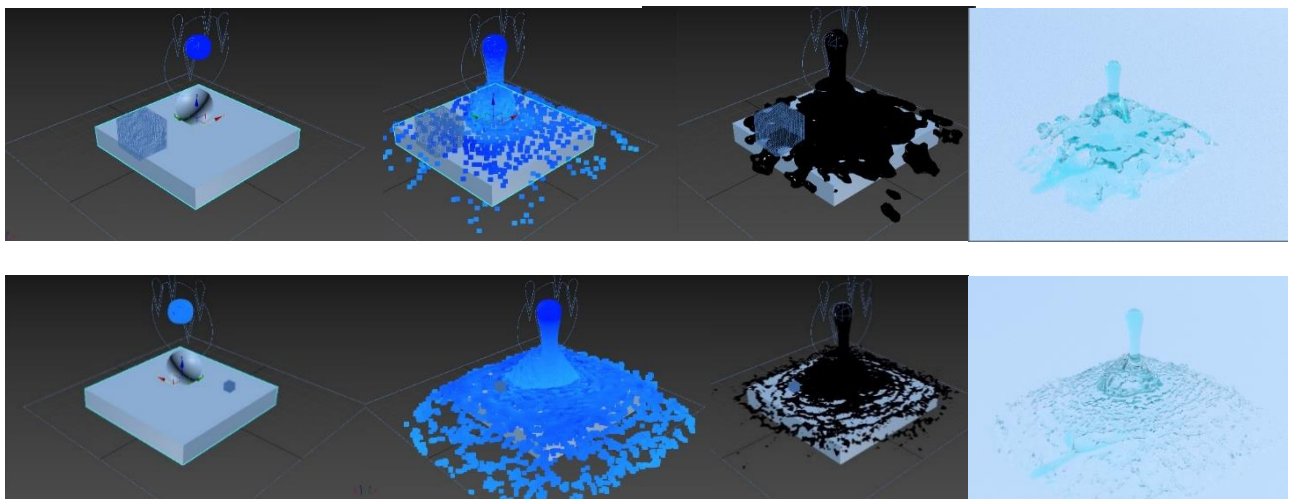


Fig. 10. Solver Grid Based, Frame 50 different voxel box volume (0.5 vs 2)

Realflow was also used to change the properties of the fluid. The focus was on three main factors: viscosity which calculates the friction between particles, resolution that calculates the number of particles for volume and internal pressure which calculates the forces that expand the fluid (see Fig.11). Furthermore, the fluid mesh has been used to create shapes, a technique often used in motion graphics to create visual designs (see Fig.12).

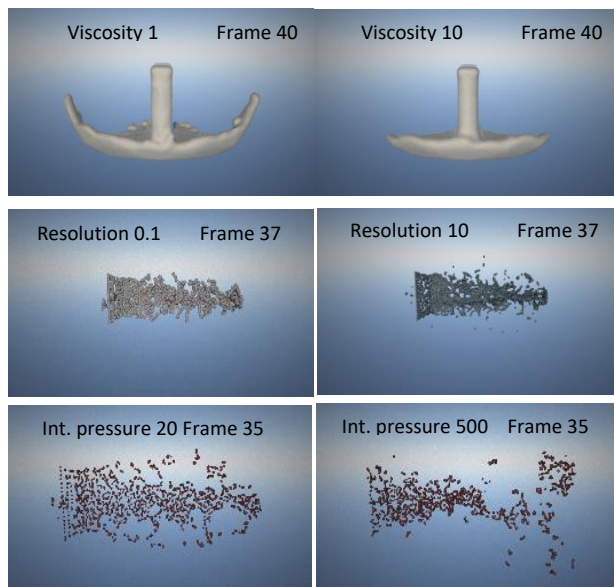


Fig.11 Changing the values in the same

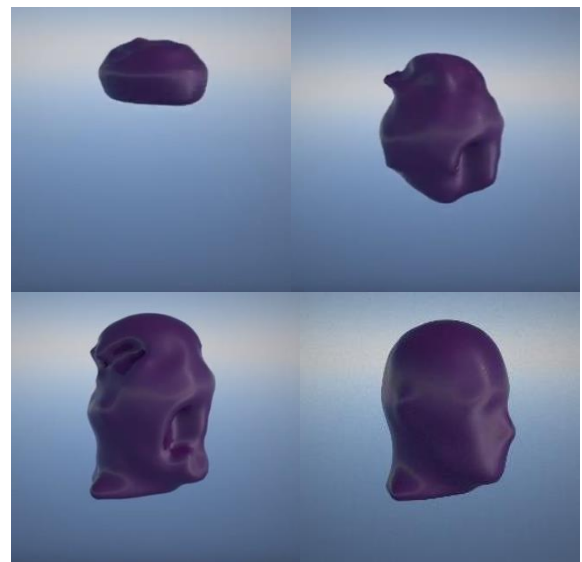


Fig.12 Creation of human face profile using a fluid mesh

Furthermore, experiments have been made to drive the simulation and follow the motion of the fluid. To achieve this, particles have been created for a low-resolution fluid in Maya. To change the motion of the fluid the particle emitter has been animated and forces have been introduced like swirl that created the turbulence which separated the fluid from its intended direction (see Fig. 13).

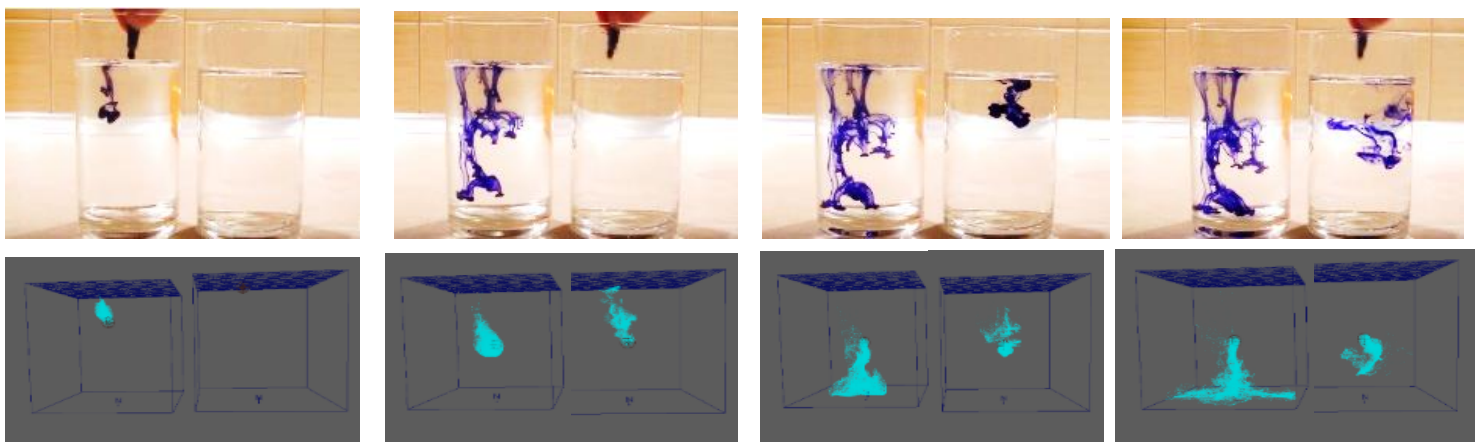


Fig.13 Experiment with particle for simulation of ink in water

3.3 Results of the experimentations

The early experimentations have been useful for the achievement of the project because it allowed to apply in practice what have been researched and to decide which path to follow. From the experimentations it resulted that the simulations need to be managed based on the shot that need to be rendered. The decision of which solver to use, needs to be taken based on the scale of the simulations and on the level of details that needs to be achieved, however simulating fluids in large scale scenes in high quality can lead to high memory and therefore to higher computer requirements.

After the experimentation it was decided to focus the project on one of the properties of the fluid field. However, it was decided that the software packages used in the experimentations would not be used and instead use a different software that allowed a better understanding of the anatomy of fluid solvers and more customisation.

4. Product

4.1 Overview of the method

The product that has been realized for the project is a system that allows to produce simulations focused on the advection of fluids. The advection is one of the main four components of the fluid field that focuses the attention on the motion and transport of the fluid (see section 2.5). To represent the advection a fluid-fluid system has been created, to show how fluids move, driven by the transport of another fluid. Three different fluids have been observed in a liquid system to see how they behave differently in the same system. Before reaching the final development of the system, many steps have been made to decide which was the best approach to take. In the next chapters the research of the product and the simulations made, are showed in detail. The product research includes the development of early prototypes made to inspect the anatomy of fluid solvers. After the examination of the solver and of the tools available in the simulations package one method has been chosen, that allowed to create a customised parameter for the manipulation of the advection of fluids.

4.2 Product research

To understand the anatomy of fluid solvers the software used was Houdini. The previous works of Peter Claes (Claes, 2009) and Jeff Lait (Lait, 2012) demonstrated how to create customised fields and fluid solvers, showing how to control and customize the simulation for specific requirements. Houdini is a node-based software which has a powerful and customizable dynamics system allowing to add features to the solver and to create large scale simulations. Houdini offers three different solvers to simulate fluids: the voxel volume (Eularian), SPH (Lagrangian), and FLIP fluid solvers (Hybrid) (see Table 2) (Side Effects Software, 2017).

Feature	Volume	FLIP	SPH
Substeps	10x	1x	100x
Bounded	Yes	Sort Of	No
Data Storage	Volumes	Particles	Particles
Control	Field	Full	Force, Velocity
Distribution	Even slices, high speed	Arbitrary slices, high speed	Arbitrary slices, good
Problems	Viscosity, Evaporation	Compression, large steps	Explosions

Table 2 comparison of fluid solvers in Houdini

In Houdini, dynamic networks are used to create simulations. Houdini has objects and solvers for different types of simulation. The most basic dynamic setup is a single object wired to a solver, with a force applied. Solvers do the work of calculating how objects behave using the data attached to objects and then used to run the simulation. If an object does not have a solver attached to the object, nothing happens when the simulations runs. Some key words need to be explained to understand the Houdini workflow (see Table 3).

DOP	Dynamic
POP	Particles
OBJ	Object
CHOPs	Channel used to control and modify any type of data
SOP	Surface inside the object, used to modify the geometry;
ROP	Render operator
\$F	Frame

Table 3. Houdini Nomenclature

The anatomy of a fluid solver has been examined in detail. This phase was useful to understand the layers of the solver and to decide which pipeline to follow to continue with the simulations.

In a FLIP Fluid dynamic system in Houdini at the top of the node chain there is an Object node, which defines the properties of the fluid (see Fig.14). The FLIP Object is connected to the solver node, after that there is a merge node and forces that are added by default such as the gravity (see Fig.15). The FLIP solver needs a geometry to emit particles from, which automatically creates the particle fluid emitter node. Also, a collision object as static rigid body needed to be add to the network (see Fig.16).

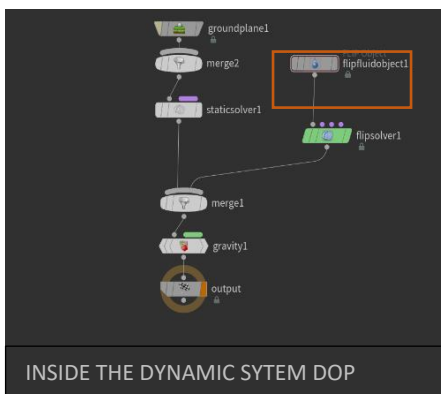


Fig.14 FLIP Object

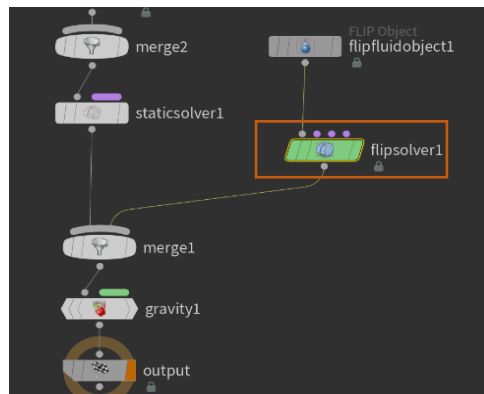


Fig. 15 FLIP Solver

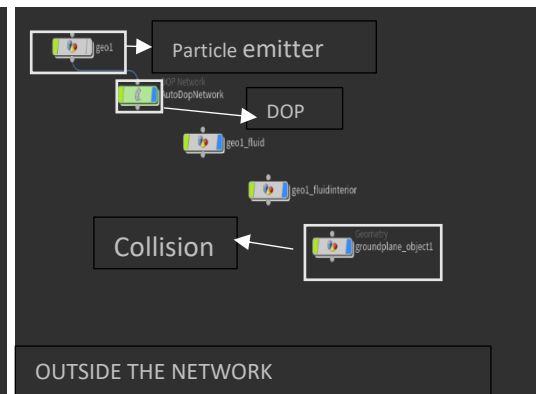


Fig. 16 Nodes attached to the network

This is the standard system that is available in the software for FLIP fluid simulations where the particles are copied into a volume grid. The Figure 17 shows the results of the FLIP dynamic network created without changing any value.

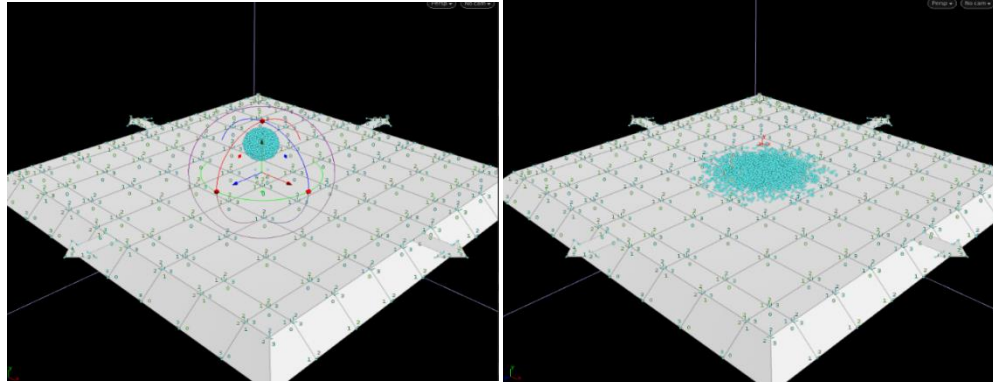


Fig.17 Results of the basic solver using a sphere as geometry

Moving forward with the product research, a solver has been also built from the beginning (see Fig.18).

- Creation of the particle system: A SOP geometry has been created, using a box connected to the node a point for volume. The node has been used to generate a field of particles creating a grid of points;
- Addition of attributes: fields have been connected to the system to add properties such as velocity, colour and acceleration;
- Withdraw of divergence: divergence is the difference between the amount of energy entering and exiting the grid. If the amount of matter entering and leaving the voxel is equal, the divergence is zero. Zero divergence is the aim in order to have mass conservation and solve the equation. This has been created by using the gas advect particle node.
- Addition of collision: the velocities values have been copied into particles and surface has been added to allow movement.

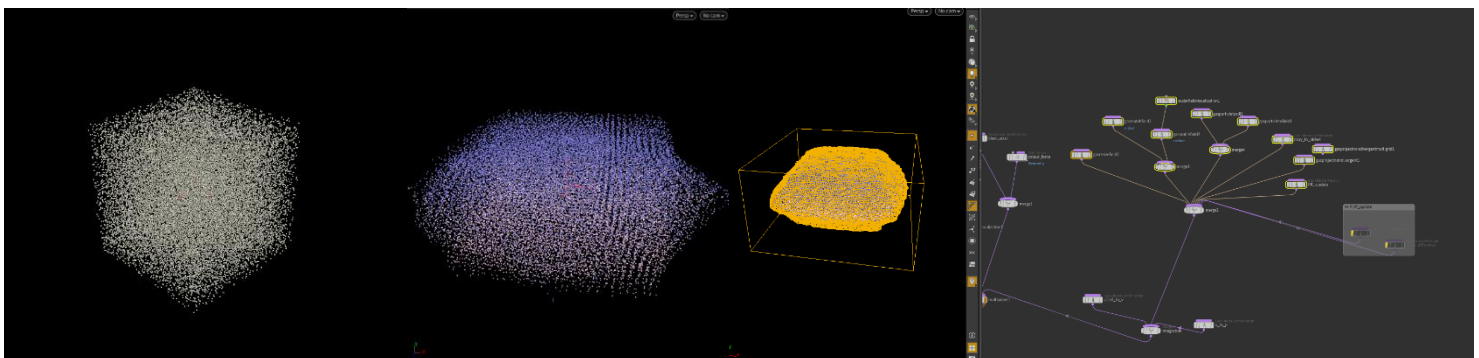


Fig 17. Building of a solver

4.3 Product simulations pipeline

The aim of the system was to develop a volumetric scene that would allow to control the motion of the simulation to recreate the advection of fluids.

To achieve this aim, a custom velocity field has been created. The system needed to represent how the fluid pushed itself around and how the substance moved with the use of motion of the surrounding fluids. The simulations have been achieved through a series of steps, which established the pipeline of the project:

1. Observation of fluids

The first step was the observation of fluids. Videos have been created to observe the motion of three fluids, glycerine, ink and milk, and used them as reference. Those fluids have been chosen because they are water-soluble meaning that water could be used as solvent to create a solution with another liquid that would allow the transport of molecules and so to observe their advection:

- The glycerine, under the form of fluid colour, is easily soluble in water because it is able to create hydrogen bonds with water molecules. Because the density of the water is heavier than the glycerine, it can quickly settle to the bottom. However, due to its high solubility, over time and with agitation it forms an aqueous solution (see Fig.18).



Fig 18. Glycerine in the water

- The ink diffuses in the water creating areas of different concentration, which is noticeable by the different shades of colour, the movement is proportional to the diffusion coefficient. Water molecules move around randomly in the mixture with the ink. As a result, the ink molecules move from areas with more molecules, higher concentration, to areas with fewer molecules, lower concentration (Balter, 2017) (see fig.19).

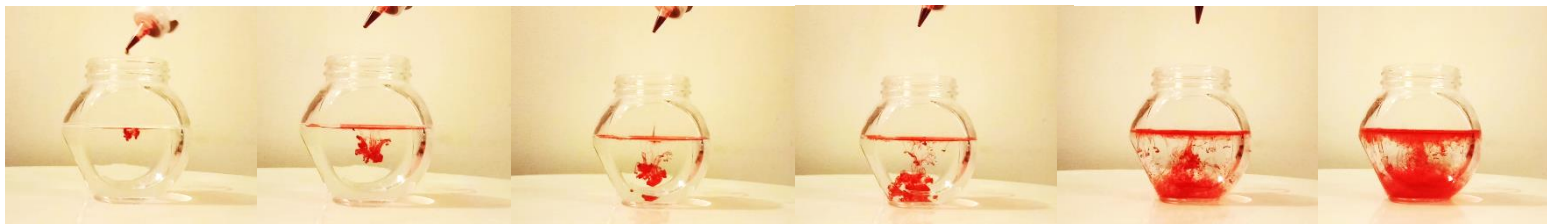


Fig 19. Ink in the water

- The milk when mixed in the water created a solution going to form a uniform liquid. To show better the turbulence the solution has been coloured.



Fig 20. Milk in the water

2. GEOMETRY

The second step in the creation of the simulations in Houdini was the creation of the geometry.

In order to achieve that, a basic sphere had to be created as a fluid source. A noise has been applied to the geometry using the mountain node. When a simulation geometry is created, Houdini creates a high-level object which is an empty object with data attached to it that the solver will need to simulate (see Fig 21).

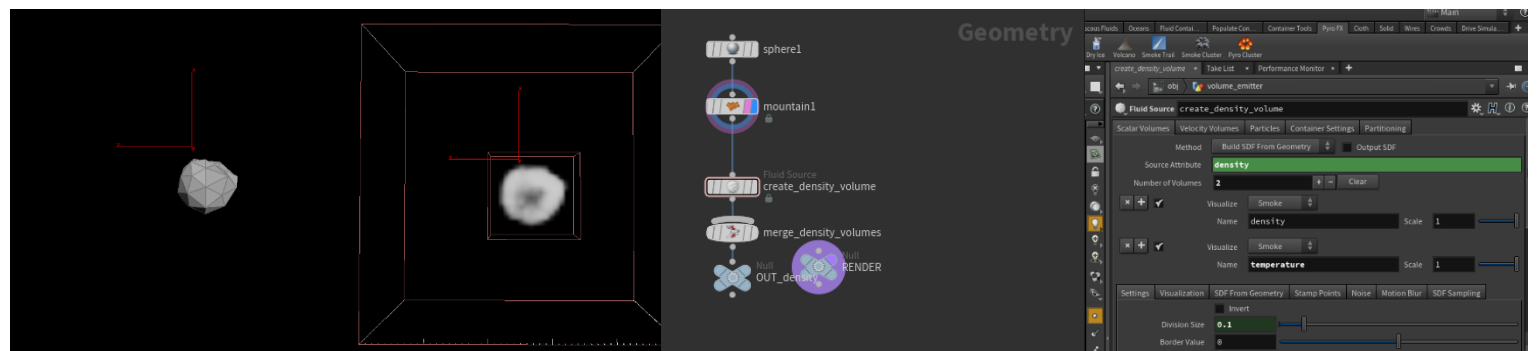


Fig 21. Creation of the sphere as fluid source connected to a mountain node

3. SOLVER

Advection effects have been created applying the pyro tool dry ice. This created the DOP network dynamic system and the solver (see Fig.22). The use of the dry ice tool allowed the flow to fall instead of rising. In addition, closed boundaries have been activated in order to create the collisions and to simulate a closed environment. This setup also allowed to add other forces such as the gravity.

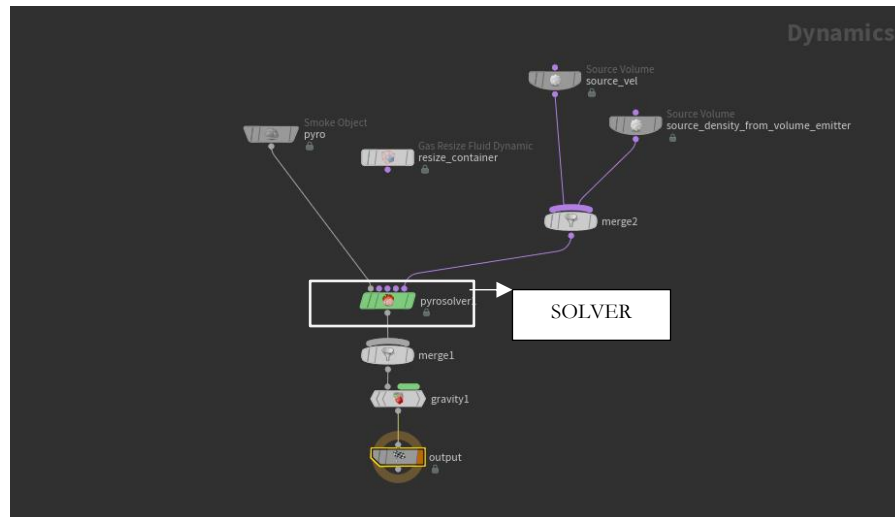


Fig 22. Creation of the solver

4.

CUSTOM VELOCITY

The most important step in the achievement of this specific simulation, is the creation of a volume that allowed to customize the velocity and to manipulate the shape of the flow. This has been done using a box filled with noises, creating a force that applies on the solver. The velocities changed based on the scale of the volume (see Fig.23).

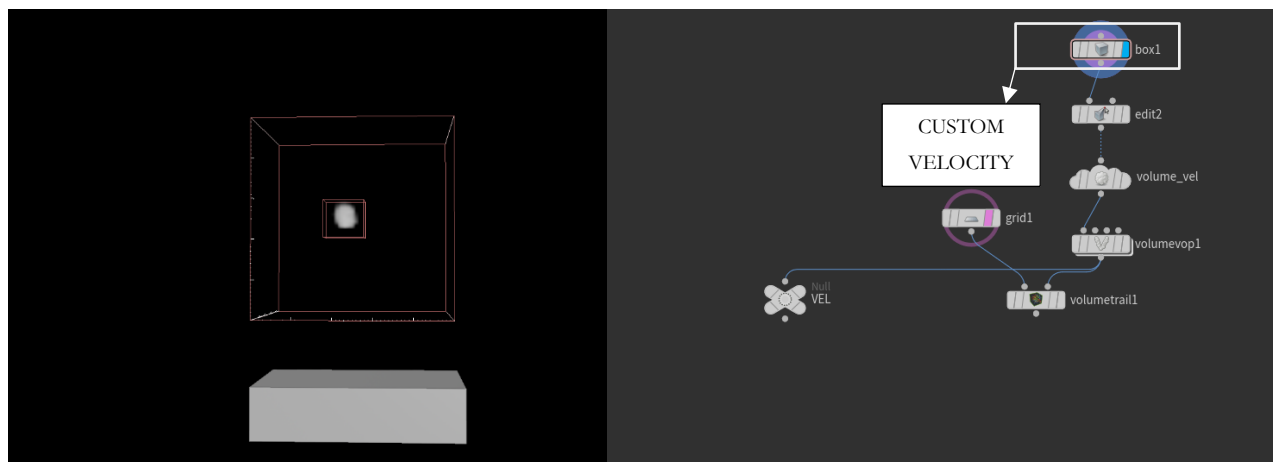


Fig 23. Creation of the custom velocity field using a box

5. VOLUME SHAPE

The setting of the volume created in the previous step has been changed to achieve the shape desired. Based on the scale of the box the simulation behaved differently, on the same timeline (see Fig 24).

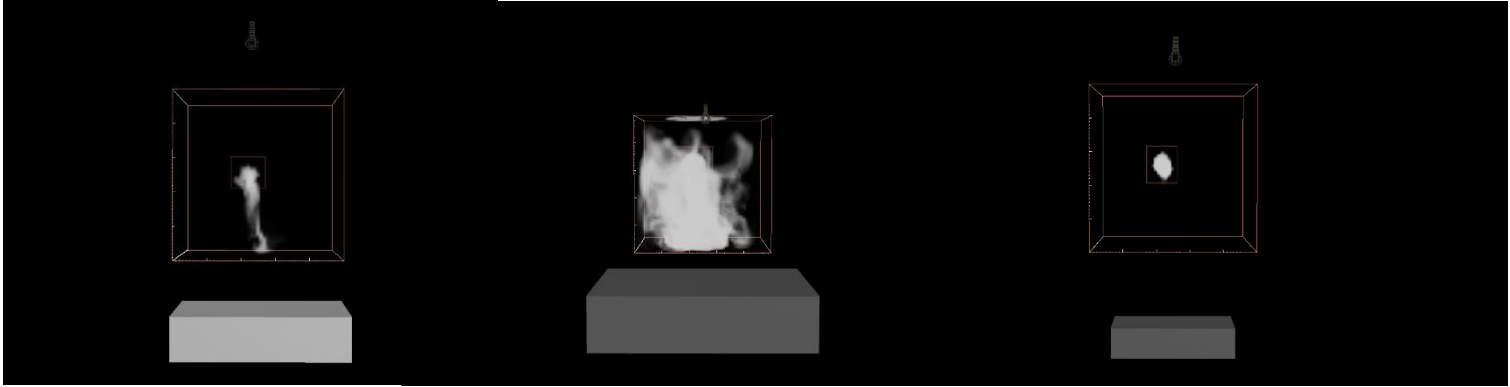


Fig 24. Comparison of the results of three different simulations at the same frame changing the custom velocity field

6. PARTICLES

After the shape desired was achieved, looking at the videos as reference, the volumes have been saved and a particle emitter has been created using the node scatter and points for volume. This method allowed other forces to be added onto the particle systems and to adjust the final motion of particles. However, the testing showed that the manipulation of the particles, was causing only minor changes in the movement of the fluid.

7. RENDERING

The strength of these simulations was not the particles but the volumes instead, because the advection of the fluid was represented achieving the shape of the volume. The rendering of the particle in Houdini would not allow to show the motion in detail, so to better represent the advection a different tool was used. The volumes have been exported as VDB files and implemented in Octane Rendering, which is a tool available as a plugin for 3D software. The 3D software used for the rendering was Cinema 4D. Every frame of the simulations has been imported separately and after changing the colour, the single frames have been rendered and exported one at the time and then used to create a sequence (see Fig 25).

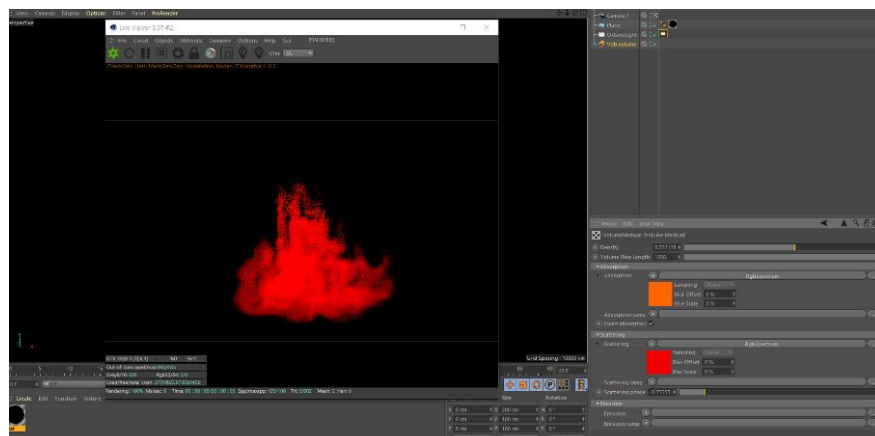


Fig 25. Rendering of the volume using Octane

4.4 Review of the pipeline

The use of Houdini allowed to understand the anatomy of volumetric simulations and to choose the best approach in creating advection. Houdini has a chain of connected nodes that allow to run the simulations through dynamic systems. The dynamic system consists of object, sources of solver data. The dynamics object holds the parameters necessary to describe the simulations, for example, when the solver sees the gravity node attached on the object, the solver takes the data to pull the sphere downwards at each time step to simulate the effect of the gravity. The main force used in the project was the custom velocity and a specific pipeline was used to achieve the simulations (see Fig 26).

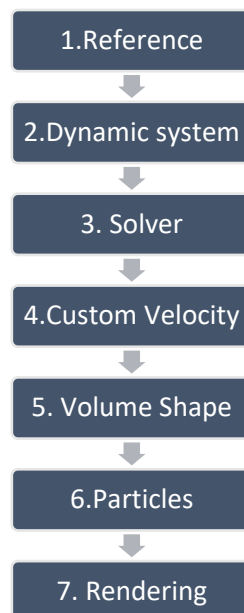


Fig 26. Summary of the steps taken for the simulations

5. Results

5.1 Analysis of the results

The simulations were performed using Houdini 12, these are the standard values of the solver created (see Table 4) and the results of the volume using the default values (see Fig.27).

	Volume geometry			Volume dynamics									
Parameters	Emitter radius	Uniform Scale	Noise	Density	Viscosity	Cooling Rate	Dissipation	Turbulence	Gravity	Custom Velocity	Velocity Scale	Velocity Blend	Impulse Activation
Default	0.5	1	1	1	0	0	0	0	-1	0	0	0	0

Table 4. Default values of the developed system

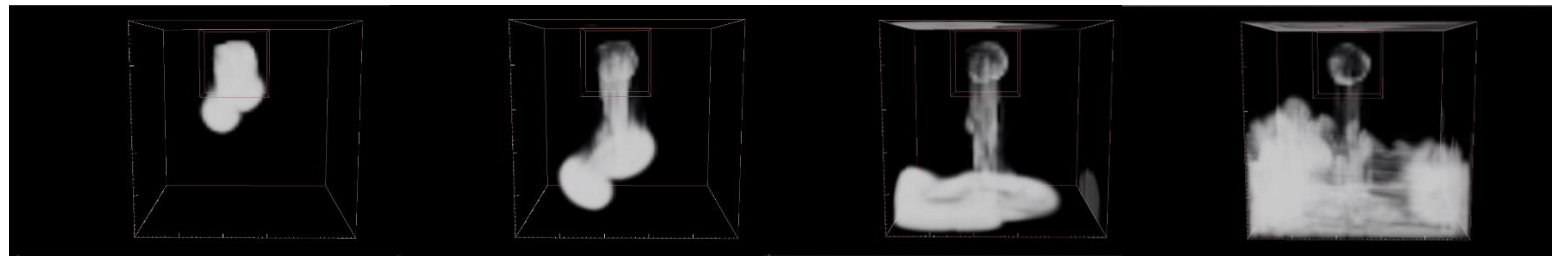


Fig 27. Result of the volume using the standard values in developed system

Other parameters are available in the software, but these have been chosen to be examined closer because they changed the look of the simulation the most:

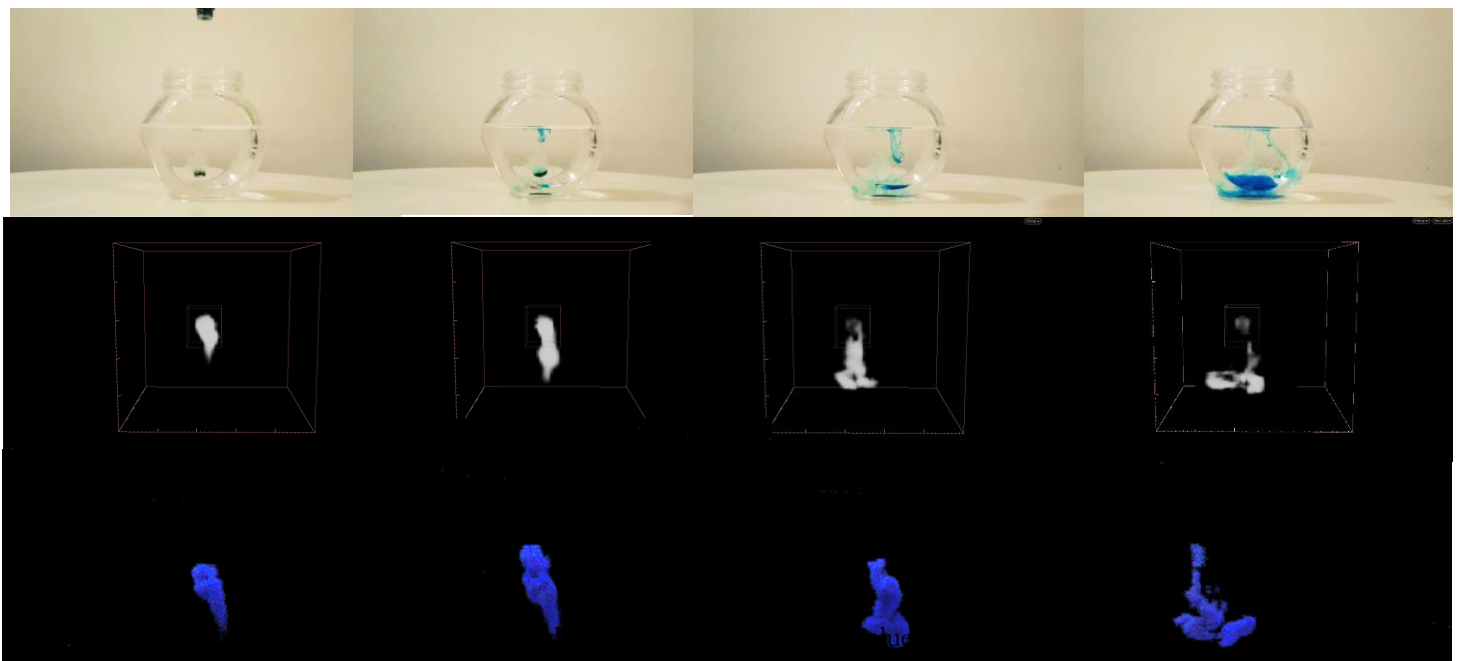
- Emitter radius: the emitter radius is referred to the geometry created in step 1. This defines the distance of the sphere from the centre of the field.
- Uniform Scale: this parameter allows to enlarge the geometry of the same amount in the three dimensions.
- Noise: the noise is given with the addition of the mountain node. This value is referred to the height parameter which indicates the displacement and makes changes in the shape of the geometry.
- Viscosity: this parameter indicates how the velocity in one grid, affects the velocity of the closer grids. High viscosity values decrease the velocity in the system, a value of 0 allows adjacent voxels to move without resistance.
- Cooling rate: this indicates how fast the temperature field goes to zero, often used in pyro simulations.
- Dissipation: it defines how fast the temperature field mixes together and it controls how smooth the fluid spreads out.

- Turbulence: it adds noise to the velocity field and it gives different characteristics to the movement of the volume.
- Gravity: it creates a gravity field where the amount of force is proportional to the object's mass.
- Custom velocity: this value is referred to the size of the box created in the step 5.
- Velocity scale: it controls the scale of the volume.
- Velocity blend: this parameter is available after applying the particles and it allows to control the amount to mix in the field velocity every sub-step.
- Impulse activation: it allows to control the activation of node, usually used along with an expression, in this case it has been used to active the start of the articles and to choose the frame.

Parameters have been changed to recreate the motion of the three fluids, using the videos as reference. The tables 5, 6 and 7 represent how the values changed, while the Fig.28, Fig.29 and Fig 30 show the visual results comparing the video, the shape of the volume and the rendering.

	Volume geometry			Volume dynamics									
Parameters	Emitter radius	Uniform Scale	Noise	Density	Viscosity	Cooling Rate	Dissipation	Turbulence	Gravity	Custom Velocity	Velocity Scale	Velocity Blend	Impulse Activation
Glycerine	0.3	1	0.59	6.44	0.5	0.835	0.006	0.2	-9	1.06	0.7	0.159	\$FF<5

Table 5. Values of the glycerine



	Volume geometry			Volume dynamics									
Parameters	Emitter radius	Uniform Scale	Noise	Density	Viscosity	Cooling Rate	Dissipation	Turbulence	Gravity	Custom Velocity	Velocity Scale	Velocity Blend	Impulse Activation
Ink	0.1	1.7	3	1.71	0.1	0.835	0.029	0.2	-9	1.50	1	0.4	\$FF<20

Table 6. Values of the ink

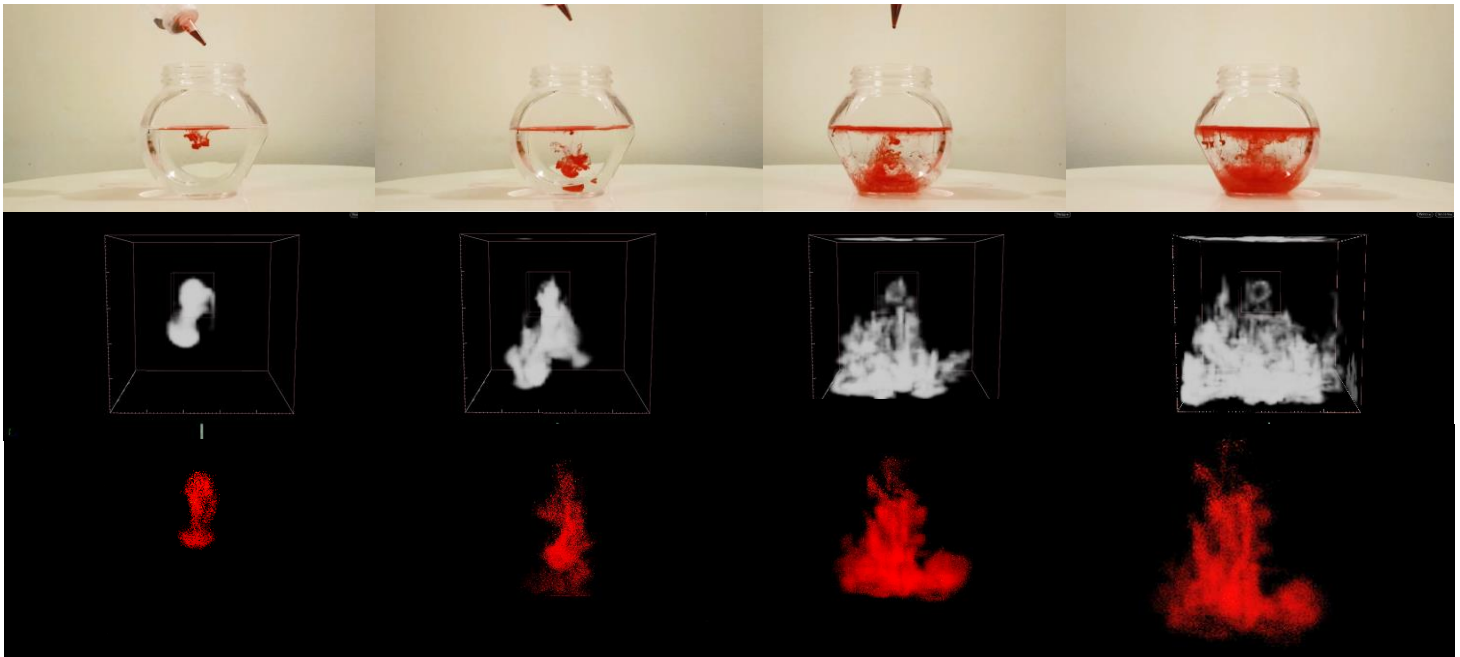


Fig 29. Result of the volume using the values of the ink

	Volume geometry			Volume dynamics									
Parameters	Emitter radius	Uniform Scale	Noise	Density	Viscosity	Cooling Rate	Dissipation	Turbulence	Gravity	Custom Velocity	Velocity Scale	Velocity Blend	Impulse Activation
Milk	0.2	1.7	0.59	1.71	0.1	0.835	0.029	0.535	-9	1	0.5	0.677	\$FF<30

Table 7. Values of the milk

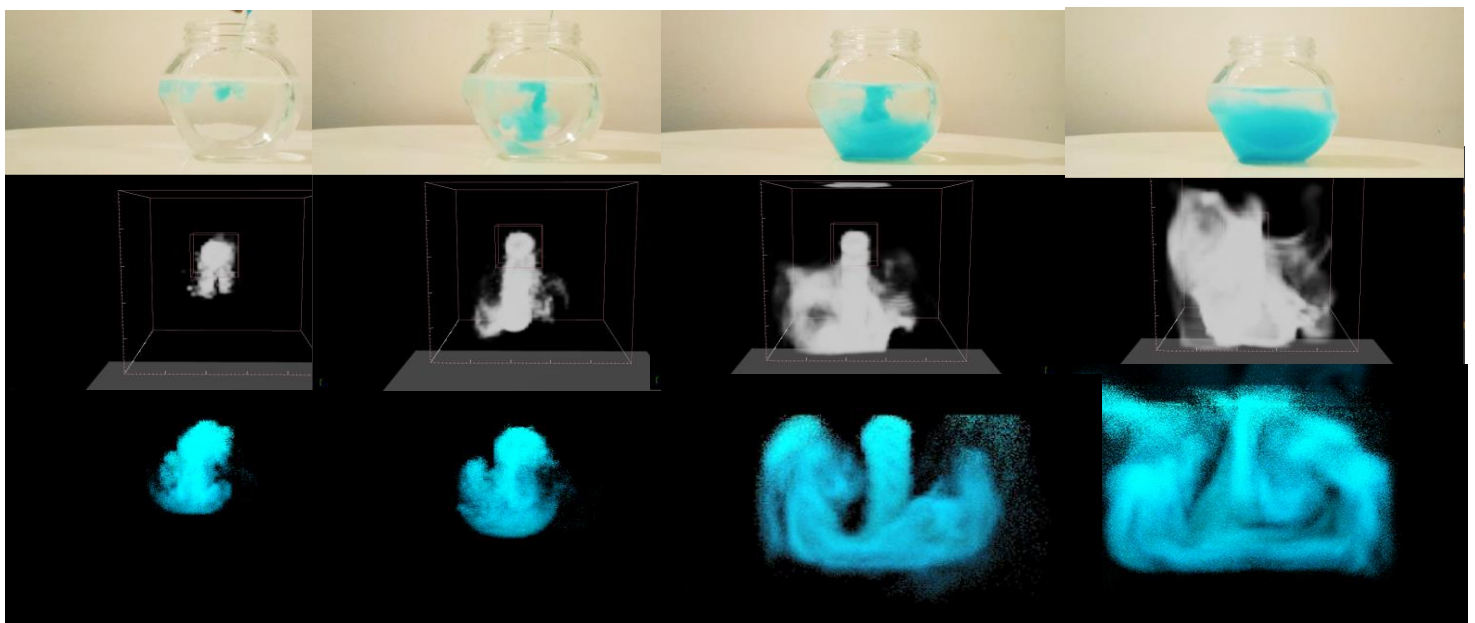


Fig 30. Result of the volume using the values of the milk

5.2 Evaluation of the results

The results of the simulations allowed to create a guidance to evaluate which parameters have been changed the most. The Table 8 represent the results of the simulations compared to one another as well as indicating which values give the significant changes in the simulation.

	Emitter size	Uniform Scale	Noise	Density	Viscosity	Cooling Rate	Dissipation	Turbulence	Gravity	Custom Velocity	Velocity Scale	Velocity Blend	Impulse Activation
Default	0.5	1	1	1	0	0	0	0	-1	0	0	0	0
Glycerine	0.3	1	0.59	6.44	0.5	0.835	0.06	0.2	-9	1.06	0.7	0.159	5
Ink	0.1	1.7	3	1.71	0.1	0.835	0.029	0.2	-9	1.5	1	0.4	20
Milk	0.2	1.7	0.59	1.71	0.1	0.835	0.029	0.535	-9	2.09	0.5	0.67	30

	Emitter size	Uniform Scale	Noise	Density	Viscosity	Cooling Rate	Dissipation	Turbulence	Gravity	Custom Velocity	Velocity Scale	Velocity Blend	Impulse Activation
Default	0.5	1	1	1	0	0	0	0	-1	0	0	0	0
Glycerine	0.3	1	0.59	6.44	0.5	0.835	0.06	0.2	-9	1.06	0.7	0.159	5
Ink	0.1	1.7	3	1.71	0.1	0.835	0.029	0.2	-9	1.5	1	0.4	20
Milk	0.2	1.7	0.59	1.71	0.1	0.835	0.029	0.535	-9	2.09	0.5	0.67	30

Table 8. Comparison of the values

The bar charts individually represent the values that changed the most which are the emitter size, noise, density, viscosity, custom velocity and impulse activation (see Fig 31).

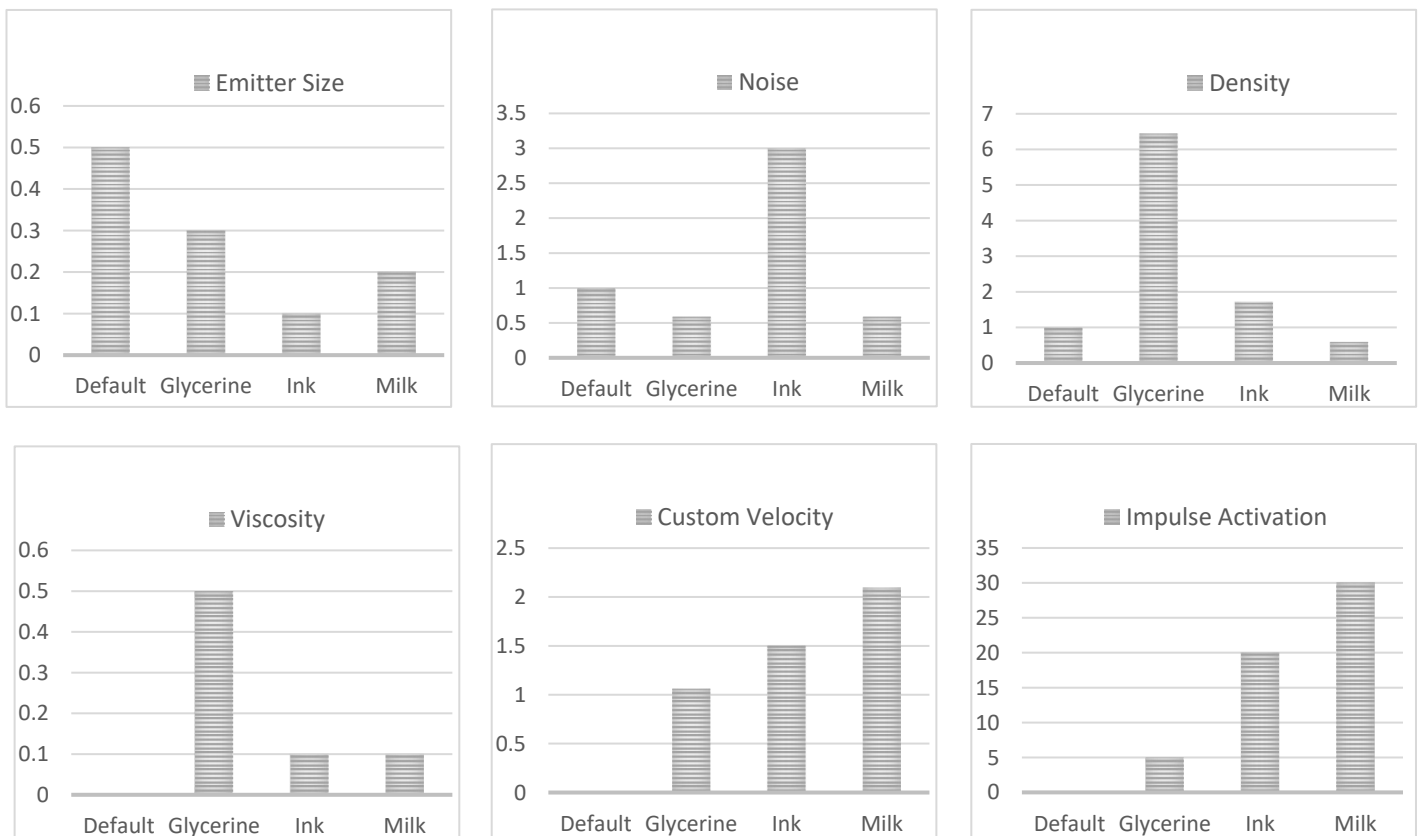


Fig 31. Individual analysis of the main parameters in the three simulations

During the simulations multiple parameters have been changed, however those parameters have been selected because they gave a different look to the motion therefore they have been used the most to recreate the advection. Comparing the charts (see Fig. 31) and analysing the values of the three fluids individually, it results that some properties show similarities, that can be also correlated to the physical world (see Table 5) and to the final renderings (see Fig.28, Fig.29 and Fig.30):

- The emitter size has a lower value in the motion of the ink. Looking at the final renderings the ink has more details in its motion, so the lower is the size of the emitter, the more details can be achieved.
- The value of the noise is higher in the ink, even though is equal in the two other fluids, the noise is useful because it allows to achieve a different visual profile.
- The density has high value in the glycerine, which also in the real world results to be the denser of the three.
- The viscosity on the other hand does not result to have relations with the real world. The glycerine is more viscous, and the ink and milk have the same values.
- The custom velocity is the parameter that provided the major changes to the visual look. As it is noticeable in the chart, there is a relevant difference between the fluids.
- The impulse activation has been an important parameter because it allowed to control the starting of the movement.

Physical world properties at 20 C^	Density kg/m3	Viscosity mPa.s
Glycerine	1261	0.950
Milk Whole	1027	2
Ink	1125	1.114

Table 5. Density and Viscosity of the three fluids in the real world

This method for the evaluation of the results was chosen because there is not a standard method that allows to evaluate fluid simulations. Professional animators and the developer of the software have been contacted to find out a way to evaluate results of simulations, but they agreed that there is not a standard guideline used in the industry, because every project is different, and it has unique requirements and objectives that need to be analysed.

6. Conclusion

The project discussed in this paper was aimed to understanding which parameters control the fluid behaviours, with a product focused on the motion and the simulation of advection, using customised fields. The simulations have been created using a 3D node-based software that allowed to understand the anatomy of fluid solvers and thus to decide which was the best approach to simulate one specific property of fluids.

The dynamics of fluids can be described by the Navier-Stokes equations and the advection is one of the property described in the equations. The advection is the phenomenon by which substances or molecules are moved by the overall motion of the fluid. To create the advection in a virtual environment, first, three different fluids have been observed in the real world. The fluids chosen were glycerine, ink and milk. They have in common the same ability to be high soluble in water, that is why they have been useful to create a system that would allow to inspect the behaviour of their motion when surrounded by another fluid. Three different scenes have been created in the software, using the same setup in order to establish the same initial environment and to manipulate the results of the simulations changing the parameters. It was important to observe the three fluids in the real world because, their behaviour has been used as main reference to achieve the look of the simulation desired.

There are many methodologies and solvers available in the animation field which allow to simulate fluids, and it is better to choose the more appropriate based on the purpose of the simulation. Therefore, it is essential to have a reference that would help to establish the type of shot or render that the artist wants to achieve at the end. From the research conducted, it has been concluded indeed that the main issues in the animations of fluids are the technologies requirements and the artist control. These issues can be partly solved by customising the parameters. The power of customisable tools has been demonstrated, using a custom velocity, which resulted to be the parameter that has been changed the most to achieve the motion desired. This parameter was not a default field available in the solver, but it has been implemented in the system with the only purpose to manipulate the motion of the fluid and therefore to simulate the advection. After the achievement of the shape desired, the simulations have been concluded and the results have been analysed. To evaluate the results, the main parameters that have been changed in the system have been compared, to understand which were the factors that allowed the simulation of the advection.

The findings of the project are useful because the pipeline and methodology used can be applied in other projects. The same pipeline can be used to represent the advection of other fluids and the evaluation of the results could be used as a guideline, even using other types of software. Furthermore, the choice to have customised fields and this specific methodology can be implemented in other type of simulations or applied to other properties of the fluid field.

The simulations could be improved by integrating more customised features, for example implementing scripting. This would allow to have a better control of the motion and to increase the

efficiency of the system. Furthermore, a depth analysis of independent proprietary tools used in industry would have helped to have a different point of view and improve the results.

A limitation of the methodology used in the project is the achievement of the photorealistic look and the level of details. The visual representation of the simulation could be improved focusing on other factors such as texture, lighting and rendering. The rendering of the particles is an aspect that has not been covered in depth, because to simulate advection the focus was more on the achievement of the shape of the volume, rather than on the rendering of the particles. The rendering has been achieved using an external tool that allowed to manipulate the volume applying colours directly on the volume.

Therefore, the implementation of the particles was not a useful step for the success of the project.

Another aspect that could be investigated further is the use of fluids secondary effects such as splash, foam, and bubbles, characteristics often inspected for simulations of water.

The main challenge of this project was the choice of the pipeline to follow, that is why the testing and the analysis of the anatomy of the software have been important steps in the production. The plans changed during the production, because the solver needed to be adapted to the aim of the simulations. Many trials have been attempted before reaching the results desired, and the discovering of the fields to manipulate was not a fast process. The methodology used has not been planned since the beginning of the project because the decisions have been made along the way, after testing and gaining knowledge of the technologies, therefore more experience with the software or similar simulation packages could help to progress with the project for future developments.

Overall, the aim of the project has been reached because the whole process helped to find out which were the parameters that manipulate the fluid as well as to have a better understanding of fluid dynamics.

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